SMART INDIA HACKATHON 2025

- **Problem Statement ID–** 25099
- **Problem Statement Title-** AI-powered monitoring of crop health, soil condition, and pest risks using multispectral/hyperspectral imaging and sensor data.
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-** 103334
- **Team Name-** X8bit

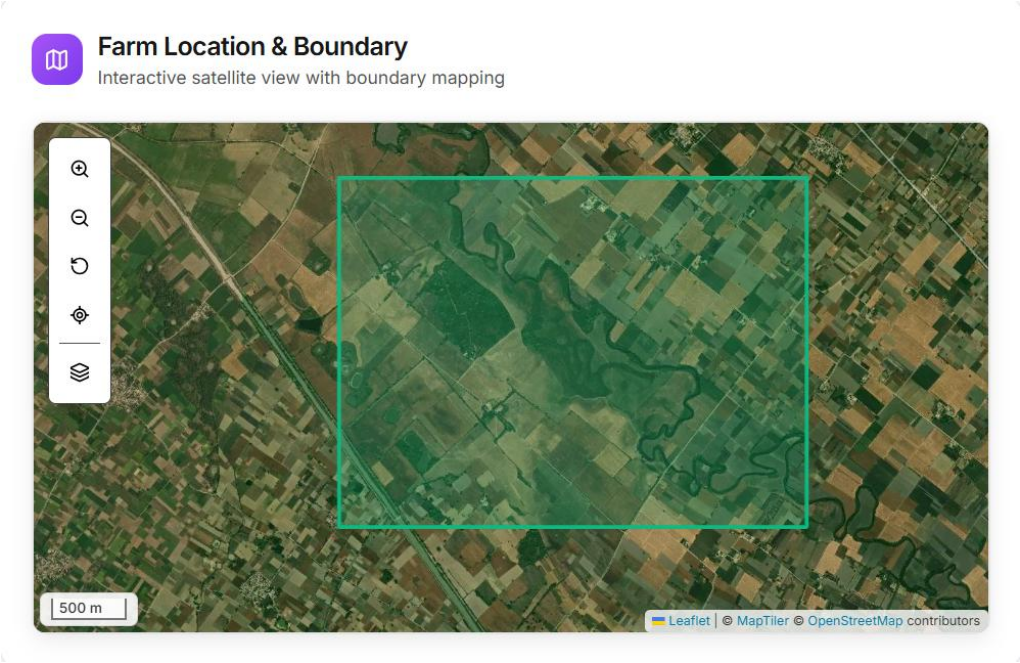
CropLab - AI Powered Crop Monitoring and Analysis System

Proposed Solution:

- **Unified Data Fusion Platform:** Ingests and aligns multi-spectral satellite imagery with historical soil data and real-time environmental data for holistic diagnostics.
- **AI Core:** Applies a combined CNN + LSTM architecture to model spatial features and predict future crop stress trends over time.
- **Actionable Zone-Specific Alerts:** Generates spectral health maps, predicted risk zones, and yield estimates.
- **Intuitive Dashboard & Mobile Alerts:** Provides Farmers, Agronomists and Researchers with clear, easily digestible insights and mobile-friendly notifications.

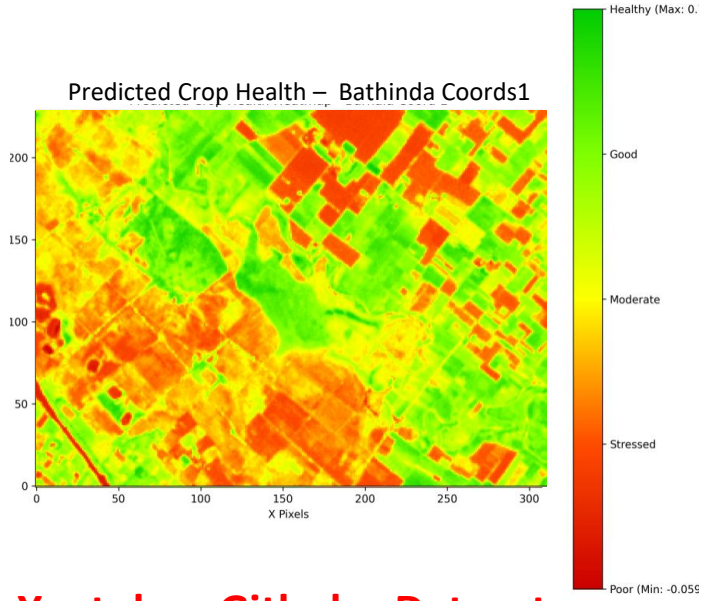
Uniqueness of Solution:

- **Fusion Model:** Uniquely combines **HUMERIS** Soil Baseline with **Sentinel-2** Imagery for comprehensive, below-the-surface context.
- **Proactive Risk Prediction:** Predicts conditions conducive to pest outbreaks and disease, enabling mitigation weeks before visible symptoms appear.
- **Regional Proof of Concept:** Solution is already trained and validated on real-world wheat crop data across the major districts of **UP & Punjab**.
- **Scalable Architecture:** Built on **MATLAB Server**, **Google Earth Engine** and MERN stack for seamless expansion across diverse geographies.



RGB Composite(from GEE)

Predicted Crop Health – Bathinda Coords1



Prototype Links: Live Project Youtube Github Datasets

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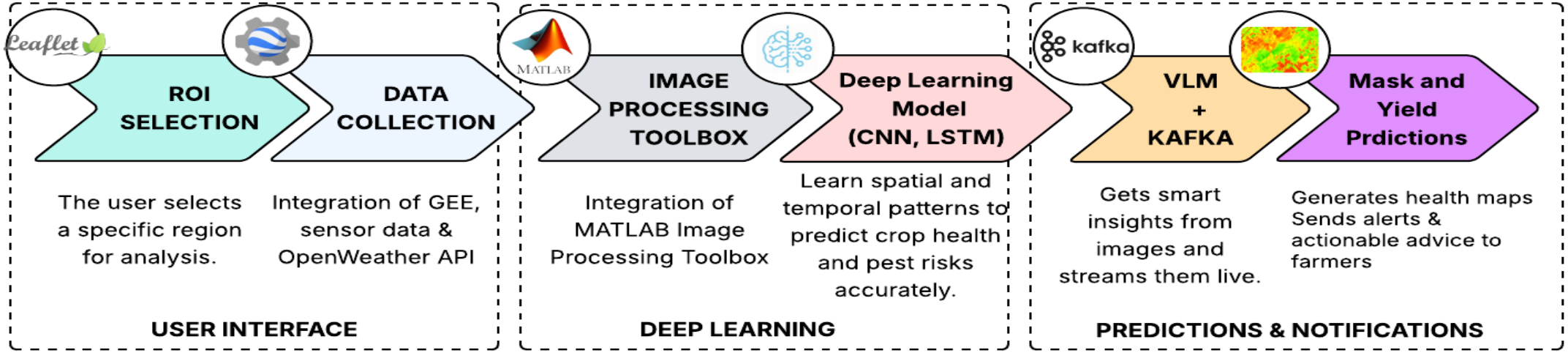
Technical Approach:

- **ROI Selection:** Users select a Region of Interest (ROI) on a Leaflet map, triggering the query for multi-spectral imagery from Google Earth Engine (GEE).
- **Data Ingestion:** Backend fetches multi-spectral image bands (B2,B3,B4,B5,B8 and B11), static soil data from HUMERIS Soil Dataset(N, P, pH, OC, ECe), and real-time environmental data (Open Weather API).
- **Data Preprocessing & Fusion:** MATLAB Production Server cleans, aligns, and integrates all data sources into a single Spatio-temporal Data cube.
- **Feature Engineering:** MATLAB extracts multiple Vegetation and Soil Indices (e.g., NDVI, EVI) necessary for advanced crop health analysis.
- **AI Core Analysis (CNN-LSTM):**
  - **CNN** performs spatial feature extraction to identify localized stress patterns.
  - **LSTM** performs temporal modeling to predict future trends like irrigation needs, nutrient deficiency, and Pest/Disease Risk.
- **Alert Generation:** The model output generates zone-specific Health Maps and Risk Masks, which trigger mobile-friendly notifications through Kafka for targeted action.

Tech Stack :

- **Frontend:**
- **Backend:**
- **AI/ML:** Matlab
- **Data Sources:**



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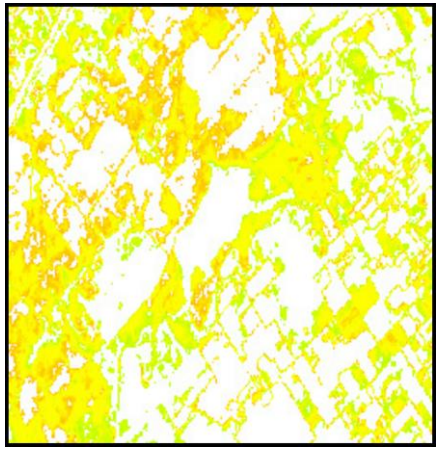
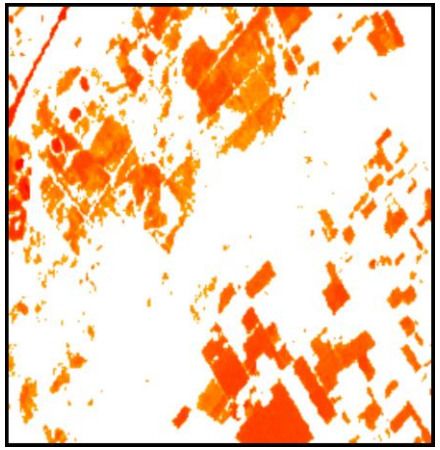
Feasibility:

- **Data Feasibility:** Integrates with **Google Earth Engine (GEE)** for continuous, high-quality multi-spectral imagery and the **HUMERIS dataset** for a robust global soil nutrient baseline.
- **Technical Validation:** Utilizes specialized **MATLAB Toolboxes** (Hyperspectral, Deep Learning) for precise spectral feature extraction and sophisticated **CNN-LSTM model training**.
- **Scalable Architecture:** Built on cloud-native data sources (GEE) and a robust backend (AWS + MATLAB Production Server), enabling fast, efficient, and large-scale deployment.
- **Practical Validation:** The AI model is **already trained and tested** on real-world yield data across major wheat-growing districts in **UP and Punjab**, confirming its operational viability.

Healthy Crop



Stressed Crop



Diseased Crop

Overcoming Challenges:

- **Data Generation:** Created custom ground-truth datasets through Python and JS scripts for enhanced model training and localized accuracy.
- **Satellite Fallback Strategy:** Employ temporal/spatial interpolation and integrate data from multiple satellites (Landsat fallback) to ensure data availability and mitigate the impact of cloud gaps.
- **Continuous Learning Loop:** Implement a process to constantly ingest new field data and farmer feedback, allowing the CNN-LSTM model to fine-tune and improve generalization for new crops and environments over time.
- **Offline Functionality:** Develop an application that uses local storage to sync recent health maps, alerts, and advice, ensuring access even with lost connectivity.
- **Model Generalization:** The Continuous Learning Loop directly addresses the need to fine-tune the model for diverse new crop types (e.g., rice, sugarcane) and varied soil profiles.

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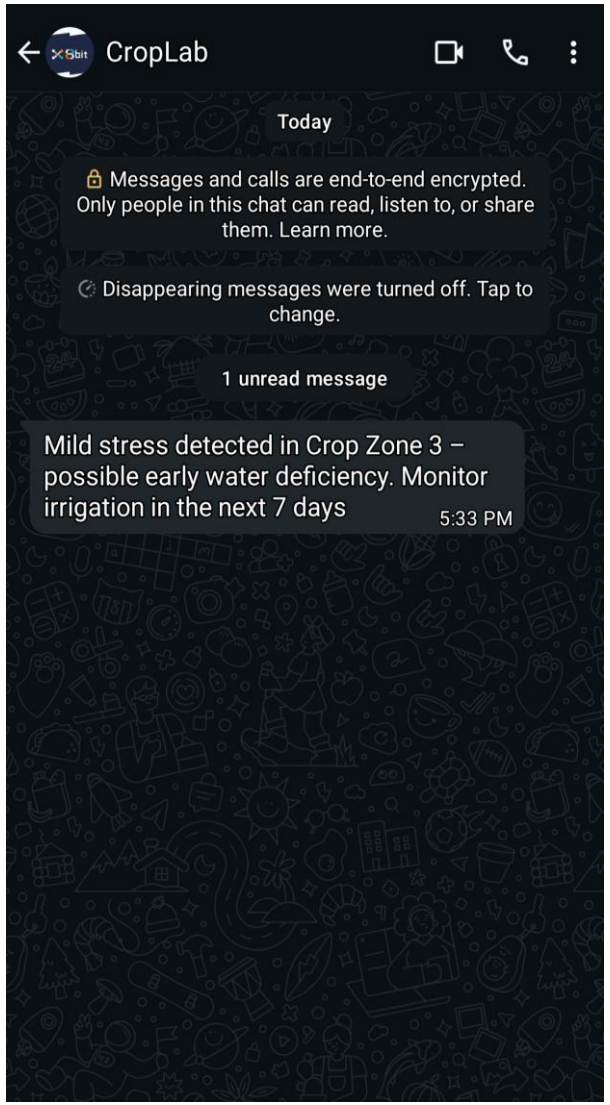
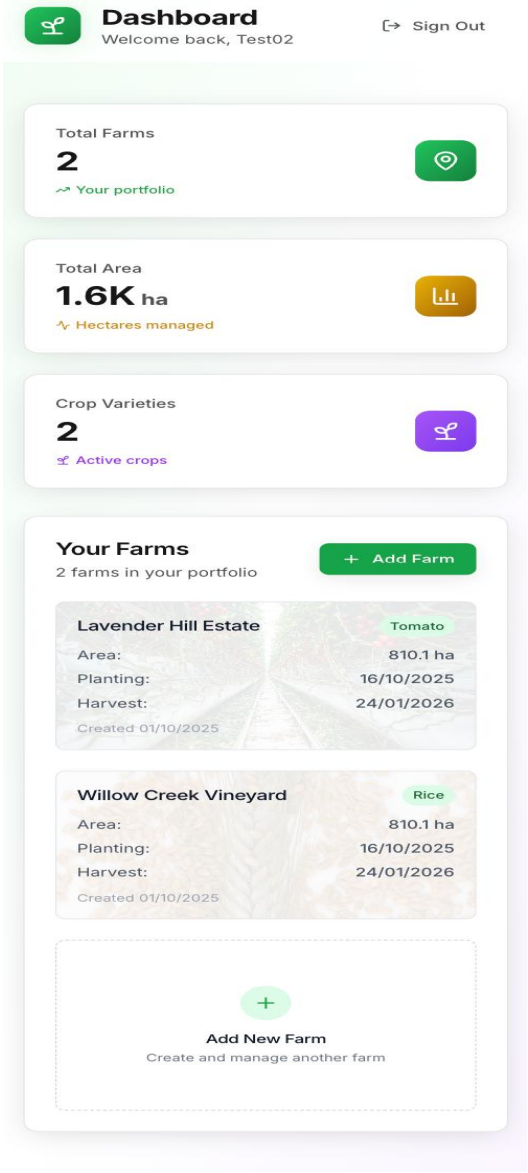
Potential Impact on Farmers

- **Crop Productivity:** Real-time crop health monitoring enables timely actions like irrigation, fertilization, and pest control.
- **Better Decision-Making:** Reliable insights on soil quality and crop conditions help plan smarter.
- **Risks Alert:** Early detection of diseases/stress reduces losses helps in higher yields and profits.
- **Less Effort & Cost:** Reduces the need for frequent manual field inspections.
- **Sustainable Farming:** Optimizes water, fertilizer, and pesticide use which helps in costs reduction.

Benefits of the Solution

- **Economic:** Increases yield and income, lowers input costs, and supports long-term farm growth.
- **Environmental:** Encourages sustainable practices, reduces chemical use, conserves water, and protects soil health.
- **Social:** Improves farmers financial security and livelihoods, significantly reduces the stress associated with unexpected crop failure, and strengthens rural community resilience.





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- **Google Earth Engine** (Sentinel-2)**:** ***Link***
- **Leaflet Maps:** ***Link***
- **MATLAB Deep Learning Toolbox :** ***Link***
- **MATLAB Hyperspectral Imaging Library for Image Processing Toolbox:** ***Link***
- **HUMERIS Global Soil Dataset:** ***Link***
- **Open Weather:** ***Link***
- ***DES Department of Agriculture and Farmers Govt. of India Dataset :*** ***Link***
- **Geojson.io** ***: Link***
- **Copernicus Browser:** ***Link***
- **Groupo De Intelligencia Computational(GIC):** ***Link***